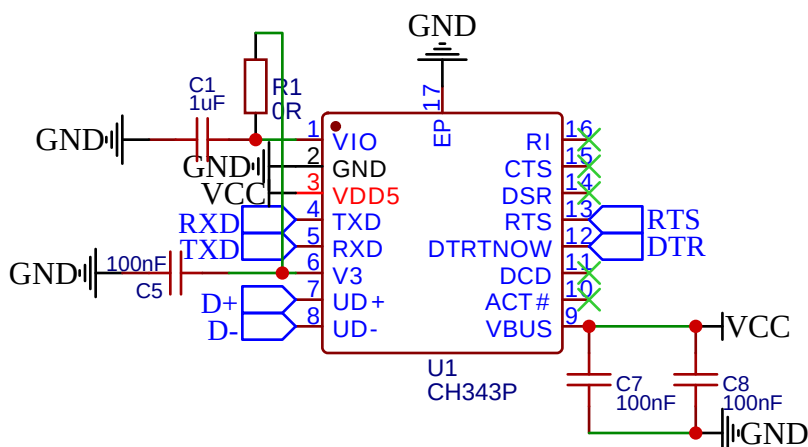


ESP32-S3

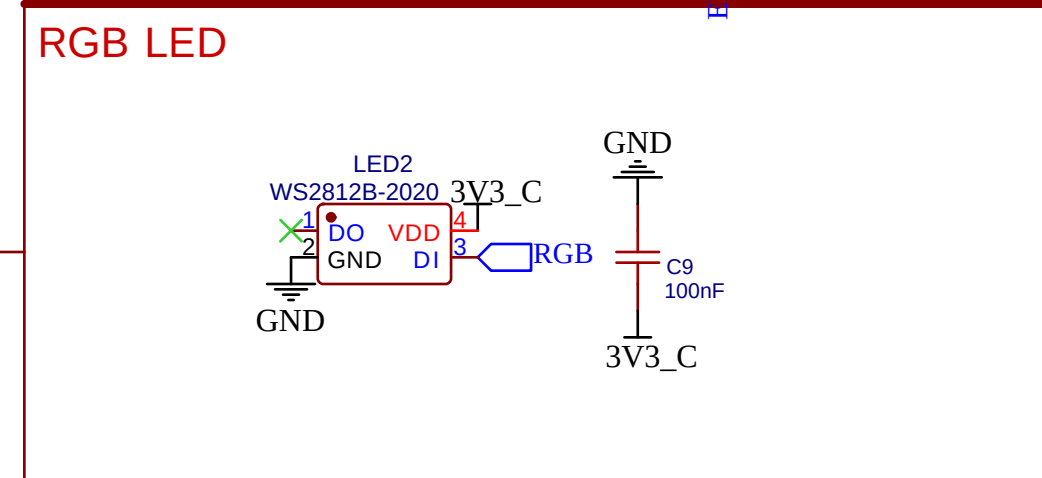
调试烧录串口

烧录USB

供电方案	串口信号电压 MCU 工作电压	VDD5 引脚	V3 引脚	V10 引脚	MCU 或外设电源
		不低于 V3 电压	额定 3.3V 左右	两者用同一电源, 1.8V~5V	
全部 USB 供电	5V	USB 供电 5V	仅外接电容	USB 供电 5V	
	3.3V	USB 供电 5V	外接电容	由 V3 供电 3.3V, 最多 10mA	
	3.3V	USB 供电 5V 经外置 LDO 电源调节器降压到 3.3V, V3 外接电容			
	1.8V~4V	USB 供电 5V	仅外接电容	USB 供电经外置 LDO 调节器降压	



供电方案	串口信号电压	VDD5 引脚	V3 引脚	V10 引脚	MCU 或外设电源
	MCU 工作电压	不低于 V3 电压	额定 3.3V 左右	两者用同一电源, 1.8V~5V	
全部 USB 供电	5V	USB 供电 5V	仅外接电容	USB 供电 5V	
	3.3V	USB 供电 5V	外接电容	由 V3 供电 3.3V, 最多 10mA	
	3.3V	USB 供电 5V	经外置 LDO 电源	调节器降压到 3.3V, V3 外接电容	
	1.8V~4V	USB 供电 5V	仅外接电容	USB 供电经外置 LDO 调节器降压	



The diagram shows four connection points. On the left, two blue vertical rectangles represent connectors for IO19 and IO20. To their right, a yellow square represents a 3V3 power supply. Further right, another yellow square represents a GND (ground) connection, indicated by a standard ground symbol below it.

滚轮编码器

The diagram shows a wheel encoder (U6, SIQ-02FVS3) connected to a 3V3_C supply and ground. The encoder's pins are configured as follows:

- Pins 4, 5, and 6 are connected to GND.
- Pins 1, 2, and 3 are connected to the 3V3_C supply.
- The encoder's output pins A, S, C, and B are connected to KEY3, KEY3, GND, and KEY2 respectively.

The 3V3_C supply is connected to a network of resistors (R16, R18, R20) and capacitors (C47, C48, C49) leading to GND.

PIN	NAME	FUNCTION
1	VOUT	Switch Output.
2	GND	Ground.
3	ILIM	Current Limit Programming Pin. Connect a resistor R_{ILIM} from this pin to GND to program the current limit. $I_{LIM} = \frac{6800}{R_{ILIM}} \text{ (A)}$
4	EN	Chip Enable Pin. Logic high to enable the device. They have integrated a 500KΩ pull-down resistor at this pin.
5	VIN	Switch Input.

电池电量检测

The diagram shows a battery level detection circuit. The battery is connected to BAT+ and BAT-O. BAT+ is connected to the non-inverting input of an op-amp (U1) through a 20K resistor (R15). The inverting input of the op-amp is connected to a voltage divider consisting of a 1K resistor (R17) and a 10K resistor (R19). The output of the op-amp is connected to the gate of a MOSFET (Q3, AO3400A). The MOSFET's source is connected to GND, and its drain is connected to the battery's negative terminal (BAT-) through a 100K resistor (R24). A 100nF capacitor (C22) is connected between the op-amp's output and the MOSFET's gate. The MOSFET's gate is also connected to the ADC_CTRL signal. The MOSFET's drain is connected to the ADC input. The battery's positive terminal (BAT+) is also connected to the ADC input through a 1K resistor (R23).

开关 & 电池接口

BAT_I

B+

GND

B-

SW1

BAT_I

SW-9*3.5*3.5MM-3P-2



